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The Impact of News Surprises on Volume- Volatility Relationship in Indian Equity Index Futures Market

Abstract

This paper empirically investigates the relationship among news releases, trading activity variables and returns volatility in equity index futures contracts traded on the National Stock Exchange (NSE), India. Using three different asymmetric GARCH models, exponential generalised auto regressive conditional heteroscedasticity (EGARCH) based and threshold auto regressive conditional heteroscedasticity (TARCH) and power auto regressive conditional heteroscedasticity (PARCH) used to test the robustness of the results. The major findings of the study are as follows: First, the estimates supports that unexpected trading activity variables have greater impact on returns volatility of index futures contracts, supporting the mixture of distribution models. Second, analyzing the effect of surprises in news announcements on trading activity variables and returns volatility, we find dividend yield, the exchange rate, put – call ratio, and the broad money (M3) significantly affects the index futures returns volatility. Lastly, the interaction effect of news announcement on volume-volatility relationship revealed significant interaction effects of surprises in news announcement variables of gross domestic product (GDP), put-call ratio, and the dividend yield.

Key words: Trading Activity Variables, News Releases, Equity Index Futures, Asymmetric GARCH

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1. Introduction

In recent years, there are evidences of growing interest in empirical research to understand the relationship among trading activities variables, macroeconomic news releases and returns-volatility dynamics of financial assets (Foster and Viswanathan, 1990; Jones et al., 1994; Lo and Wang, 2000; Tauchen and Pitts, 1983). Though, the empirical research was mainly focused towards industrially developed markets *for e.g.* U.S. capital markets with limited evidences in context to emerging markets like India. In this paper, we primarily aim to bridge the existing gap in the literature by examining investors' reaction aftermath surprises in news announcements especially policy oriented and idiosyncratic ones and its impact on the asset return volatility, and uncover new results in trading volume variables-volatility relationship in equity index futures contract (CNX S&P Nifty 50 index futures) of National Stock Exchange (NSE), India.

The study is even more imperative as some of the emerging market nations have witnessed phenomenal growth, post implementation of major structural and economic reforms, which resulted the financial markets to grow significantly. For instance, China is ranked second only to US in stock market capitalization, even surpassing that of the European Union and Japan; India, foremost to have a recognized exchange in Asia (Bombay Stock Exchange, BSE), followed by the development of a fully electronically traded exchange, National Stock Exchange (NSE), emerged as the largest exchange in India and one among the largest exchanges in the world with approximately INR 300 Billion¹ as daily turnover of future and option (F&O) segment. Rapidly

¹Information retrieved from the website of NSE, India

growing capital markets characterised by excess volatility has drawn our interest particularly to index futures contracts to explore, whether the volatility patterns in emerging markets are different from that of the developed economies.

Most of the extant studies have focused their interest on studying asset return volatility and information arrivals, which remained the building block for “Mixture of Distribution Models” (MODM) (Clark, 1973; Epps and Epps, 1976; Foster and Viswanathan, 1995), which posits that the volume-volatility relationship originates from a joint dependence on a underlying common event or information flow variable, and/or “Sequential Information Arrival Models” (SIAM) (Copeland, 1976; Jennings and Barry, 1983), which otherwise, suggests a lead-lag (casual) relationship between volume and volatility, also supported by noise trading hypothesis (Iori, 2002 ; Harris and Raviv, 1993). Many of the these previous studies based on one of the theoretical models documented positive contemporaneous correlation between volume and volatility in equity and futures markets (Gallant et al., 1992; Chen et al., 2001; Ciner, 2002), widely using volume as proxy for information arrival.

Though earlier studies measure trading activity based on volume alone, however, Bessembinder and Senguin, (1993), was first to develop a model to investigate the price volatility and trading activity variables (trading volume and open interest) relationship, where open interest serving as a measure for market depth, since it reflects the order flows in the futures transaction and willingness of the market participants to open new position in the presence of price volatility. In consistent with previous studies, Bessembinder and Seguin (1993), documented strong positive relation between contemporaneous price volatility and volume, while Diagler and Wiley (1999) and Watanbe (2001) in related type of study obtained similar findings to those of Bessembinder and Seguin (1993).

Following the method developed by Bessembinder and Seguin (1993), Diabler and Wiley(1999) and Watanabe (2001), we investigate the relationship between price volatility, trading activity (trading volume and open interest) in the Indian future index market of National Stock Exchange (NSE), India. For testing, the above relationship, we used asymmetric GARCH models in consistent with Gulen and Mayhew (2000), Lamoureux and Lastrapes (1990) to investigate the price volatility-volume relationship in index futures market, employing daily trading volume as a measure of information flow (the rate of information arrivals) based Clark's (1973) mixture of distribution model(MODM).

Furthermore, taking into account evidences from related asset markets, such as equities (Koutmos and Booth, 1995; Bernanke and Kutter, 2005), we examine whether or not the equity index futures respond asymmetrically to scheduled news announcements, especially to the surprises component of the news announcements. We contend that the study of how equity index futures respond to positive versus negative news surprises would not only be informative from the perspective of efficient market theory, but also settle some of the dichotomy of previous works reporting mixed or relatively weak link between news announcements and asset prices.

Using comprehensive data set for a period of 5 years from January 2, 2012 to December 29, 2016 for scheduled economic, idiosyncratic news announcements and forecasts, we calculated the news surprises in the announcements. These data, along with the price information of S&P Nifty 50 index futures, allow us to determine which announcements significantly affect the returns volatility. As the data set also contains information about the trading activity variables which enables us, to further investigate the effects of different announcements on trading activity and the interactive role of news surprises on volume-volatility relationship in futures market.

As Andersen's (1996) posited that different types of news have different frequencies of arrivals and have varying impact on pricing behaviour. Taking into account that finance literature has identified a dozen of such different news releases that have significant effect on financial markets (Andersen et al. 2003), as a part of the study, we examined eight different news releases in our analysis released on different dates and at different frequencies. The releases are as follows: Gross domestic product (GDP), wholesale price index (WPI), broad money (M3), 91 day Treasury bill rate and Mumbai Interbank Offer Rate (MIBOR), dollar-rupee exchange rate, put-call ratio(sentiment indicator), and dividend yield. Using, Exponential Generalized Auto-Regressive Heteroskedasticity framework (EGARCH), we aim to identify those announcements that significantly affect realised volatility by incorporating positive and negative news surprises in the conditional variance equation to test significant impact of these announcements on the returns volatility, volume and volume-volatility relationship of Nifty 50 equity indexed contracts of NSE. Next, we check the robustness of our results using alternative specification.

Though the previous studies, have documented news effect on stock (Bernanke and Kutter, 2005; Boyd et al. 2005); on bonds by Nowak et al. (2011); on exchange rates by Chen and Gau (2010). However, there are little evidences to evaluate the effects of news releases on futures markets in emerging market settings like India. The present paper is an attempt to examine the responsiveness of several market activity variables including return, volatility and trading activity variables. Our study is comprehensive in scope connecting literature from different market segments, to examine the impact of "*news announcements*" on security prices, returns volatility and volume-volatility relationship. While each of these topics has been studied separately in the literature, the connection between them has received partial attention. Finally, the analysis of the study has theoretical implications, theoretically volatility is key component in

derivative pricing and understanding the response of volatility to news shocks would allow to better price derivative contracts.

Major findings of our study are as follows: we find that the expected and unexpected components of trading volume and open interest affect returns volatility significantly and that the unexpected components of volume and open interest affects returns volatility more than the expected components demonstrating that unexpected trading volume and open interest contains more information in explaining returns volatility. In this paper, we also analysed the importance of news announcements on trading activity variables and returns volatility, we find the following of the news announcements affects the index futures returns volatility significantly: dividend yield, the exchange rate, put –call ratio, and the broad money (M3). Finally, we used the product of the surprises in news announcements and the trading activity variables to estimate the interaction effect of news announcement on volume-volatility relationship; significant interaction effects are found with news announcement variables of gross domestic product (GDP), put-call ratio, and the dividend yield, thus receiving support to some of our hypothesis.

The remainder of this paper proceeds as follows. Section 2 the literature review and hypothesis are presented. Section 3, we describe the data and the estimated EGARCH framework to analyze the data. In section 4, we present the empirical results whether expected and unexpected components of trading volume and open interest are related to conditional volatility and whether surprises in news announcements affects the returns volatility and the volume-volatility relationship. In section 5 we present the results of the robustness checks using alternative specification. And in section 6, the conclusions are presented.

2. Literature Review and Hypothesis

Several theoretical models attempted to relate price movements, trading volume with information assimilation. For instance, the mixture of distribution models (MDH) relies on news to explain asset returns (Clark, 1976; Epps and Epps, 1976; Tauchen and Pitts, 1983; Harris, 1986). As new and unexpected information arrives, traders revise their expectation about the future price, resulting in both contemporaneous change in future price and trading volume to change in new information arrival. While Copeland (1976) and Jennings and Barry (1983), “sequential arrival of information model” posits that sequential arrival of news in markets results in volume-price movements through information shocks and the model supports positive association between trading volume and price volatility, and that the price volatility is predictable through trading volume knowledge. In addition, Blume et al., (1994) argued that trading volume can provide distinct information from that of price as volume captures the important information on the quality and/or precision of traders information signals.

Numerous studies by Epps and Epps (1976), Harris (1986), Gallant et al., (1992) and Ciner (2002) have tested the price- volume relationship using data samples from both equity and futures markets and found empirical support of positive relationship between price volatility and volume in consistent with theoretical models developed by Clarks (1973), Copeland (1976) and Blume et al., (1994). While Lamoureux and Lastrapes (1990) examined the efficacy of autoregressive conditional heteroscedasticity (ARCH) effects using GARCH specification by incorporation trading volume as a proxy for information variable into conditional variance equation. The result of the study supports that the information variable explains much of the GARCH effects. Whereas Foster (1995), unlike Lamoureux and Lastrapes (1990), posited that GARCH effects remain by including current volume in the conditional variance equation, which is consistent with MODH.

Bessembinder and Senguin (1993) studying eight futures market provided empirical evidences of positive relationship between futures return volatility and trading activity and in particular the influence of unexpected volume shocks contributing more to price volatility than the expected volume. Further, they reported that expected open interest is negatively related to price volatility in futures market, indicating that increased market depth or liquidity mitigates price volatility, thus supporting market depth theory (Kyle, 1985, Admati and Pfleiderer, 1988).

Ragunathan and Peker (1997) investigated the futures contracts traded on the Sydney Future Exchanges, while Diagler and Wiley(1999) examined futures contracts traded on Chicago Board of Trade (CBOT), obtained analytical results that confirm the empirical findings of Bessembinder and Seguin (1993). Fung and Patterson (2001) investigated the volume–volatility relationship in currency futures market. The results report negative relationship between market depth and volatility suggesting that increased market depth reduces price volatility. While related study by Frino et al. (2012) and Hong and Yogo(2012), reported similar type results that of Bessembinder and Seguin (1993) and Diagler and Wiley (1999), they further argued that movements of open interest may be more informative, than prices in presence of hedging demand and lesser risk absorption capacity of futures markets.

In view, of the above discussion, it is apparent that trading activity variable of volume and open interest are important determinant of futures price volatility. As such, we investigate the relationship in emerging market to provide additional evidence by empirically assessing CNX S&P Nifty 50 equity index futures contract of NSE, India.

And hence in this study, we test the following hypothesis:

H1: Trading activity variables of expected volume is positively related with returns volatility

H2: Trading activity variables of unexpected volume is positively related with returns volatility

H3: Trading activity variable of expected open interest (OI) is negatively related with returns volatility

H4: Trading activity variable of unexpected open interest (OI) is positively related with returns volatility

Extant studies tend to support significant effect on asset pricing and asymmetric return volatility response to macroeconomic news releases. For instance, Chen et al. (1986), investigate the systematic influences of economic state variables on stock market returns and on asset pricing. Identifying five potential factors, they conclude that industrial production, default risk premium, term structure spread emerges as the dominant one and somewhat weaker evidence was found in support of unanticipated inflation affecting asset pricing. Studies by Fama, (1981), and Pearce and Roley (1985) examine the affect of inflation and money supply on stock returns. They posited that stock returns to be negatively related to inflation and to money growth. However Chan et al. (1998) mitigated the role of macroeconomic factors exception being default risk and term premia; further fundamental attributes such as firm size, book-to-market ratio and dividend yield are also set to influence assets returns and returns co-variation.

In a related study Flannery and Protopapadakis (2002), using an extensive dataset provided additional insights that macroeconomic variables affect both equity returns and conditional volatility using GARCH model on daily equity returns. They identified three nominal variables (CPI, PPI and Monetary Aggregate) and three real variables (Employment Report, Balance of Trade, and Housing Starts) as strong contenders of market risk factors. Similar, results was also supported by Adams et al. (2004), that unanticipated inflationary news has notable effects intra-day equity returns. However Andersen et al. (2000) studying the effect of Japanese macroeconomic news announcements on intraday volatility; they found scant evidence to

explain any systematic rise in intraday volatility by the release of macroeconomic news announcements.

Using GARCH models on U.S. monthly equity returns Hamilton and Susmel (1994) tested whether switching from low volatility to high volatility regime is depended on broader economic conditions. They showed that macro conditions significantly affect equity returns and returns volatility and the volatility remains high during recessionary periods. Errunzaanad Hogan (1998) document using VAR models for European stock returns for the period of 1959-1993, that money supply volatility Granger causes returns volatility in Germany and France, and that industrial production Granger causes returns volatility in Italy and Netherlands. While Gospodinov and Jamali (2015), studied the dynamic response of stock market volatility to monetary policy shocks using bivariate VAR-GARCH model, they results of the study supported asymmetric return volatility response to policy shock.

Several existing papers, have investigated the link between asset pricing and macroeconomic news announcements in other markets. For instance, Fleming and Remolona (1999) studying price volatility and trading behaviour for U.S. Treasury securities concluded that announcements of PPI, CPI and employment report affects volatility, whereas similar study by Balduzzi et al. (2001) with a broader set of macroeconomic announcements, documented that significantly and persistently increases volatility and trading volume post news announcements and they related price changes to the surprise component of the announcement unlike Fleming and Remolona(1999). While Green (2004), studied impact of trading on the prices of 5 year US Treasury noted around scheduled macroeconomic releases and found significant informational role of trading following the macroeconomic announcements, whereas stronger evidence was provided by McCredie et al. (2014), they observed that monetary policy announcements and explanatory minute releases significantly affect implied yield and volatility of Australian interest

rate futures contracts. Study by Evans and Lyons (2005) on currency markets presented theoretical and empirical evidences that asset prices are affected by news announcement and that the response to news is not instantaneous rather takes times to get incorporated in asset prices.

In this study, we use macroeconomic and idiosyncratic news announcements as a proxy for informational arrivals, as security prices are affected by the adjustments in expectations to the changing economic conditions driven by surprises in news announcements of GDP, inflationary rate, money supply and among others. And that previous studies have shown that news announcements are linked to changes in security prices. Bernanke and Kuttner (2005) analyzed stock markets, while Balduzzi et al. (2001) analyzed the bond markets. However the existing studies are limited to fewer asset classes, while in reality, news affects prices of other asset classes in different markets. This provides an opportunity to investigate how news announcements and surprises in news announcements contribute to pricing, returns volatility and affect the volume-volatility relationship in futures market. And hence, we test the following hypothesis:

H5: News surprise announcement affects returns volatility and the magnitude of negative news surprise is different from that of positive news surprises

H6: Negative news surprises affect the returns volatility more than that of positive news surprises

H7: News surprises moderates the relationship between trading activity variables (volume and open interest) and returns volatility

3. Data and Methodology

For the current study, we have used times series data of Nifty 50 futures contract as traded at National Stock Exchange (NSE), India from 03 January, 2011 to 29 December, 2016. The data contains daily closing prices, the last transaction price, time to expiration and in addition trading volume and open interest. Following, Bessembinder and Seguin (1993) and Chen and Tai (2014), the index futures prices are taken from the nearby month contracts usually are the most actively traded. Further, we roll over the nearby contract into the second-nearest contract on the expiration date for continuity. And the return of the future price series R_t is computed by taking the natural logarithmic difference in the price levels.

$$R_t = 100 \times (\ln P_t - \ln P_{t-1}) \quad (1)$$

where P_t is the closing price of Nifty 50 futures contract on date t . And, other time series variables are obtained from handbook of statistics of Reserve Bank of India (RBI), Bloomberg database, and from economic outlook of Centre of Monitoring of Indian economy (CMIE).

Table 1 lists some the basic descriptive statistics of the returns, trading volume and open interest of the Nifty 50 futures contract. The mean daily return on Nifty 50 futures is 0.00019 with returns standard deviation 0.00436 per day. The negative sample skewness (-0.29603) indicates that the returns series is partially negatively skewed, whereas the kurtosis value of 5.07147 is an indication that the unconditional distribution of the returns exhibits fat tails and excess peakedness against the normal distribution, which also is also confirmed by the JB statistics; while the corresponding mean and standard deviation of log-volume (log-open interest) are 5.90511 (7.23971) and 0.17932(0.12052) with skewness -1.69247 (-0.90327) and kurtosis 12.8613(4.81357).

<Insert figure1 about here>

<Insert table 1 about here>

3.1 Partitioning Expected and Unexpected Volume Variables

Following Schwert (1990), Bessembinder and Seguin (1993), Daigler and Wiley(1999) and Tai et al. (2011), the trading volume variables are partitioned into expected and unexpected components using ARIMA²(p, q) procedure, in order to investigate to what extent surprises versus trend activity variables affect the volume –volatility relationship. And for the empirical study, we have taken logarithmic volume series against the realized values for analysis. The expected component is the predicted/fitted value from the ARMA, while the unexpected (residual) value is the total volume minus the expected volume. The ARMA (p, q) model is specified as follows

$$TA_t = \alpha_0 + \sum_{i=1}^m \alpha_i TA_{t-i} + \sum_{j=1}^n \beta_j \varepsilon_{t-j} + \varepsilon_t \quad (2)$$

where, TA_t represents the total trading volume (V_t) and open interest (OI_t), ε_{it} is the residual in the above equation. Prior to decomposition, the presence of unit roots in volume and open interest is also tested using Augmented Dickey-Fuller and Philips- Perrontests. The findings of the test as reported in table 2 firmly rejects the null hypothesis of unit root at the significance level of 1% for both the volume series necessary a condition to avoid spurious regression.

<Insert table 2 about here>

²Before decomposition of the trading activity variables into fitted and residual components, the stationarity condition is tested for both the trading activity series using Augmented Dickey Fuller (ADF) test, and Philip-Perron (PP) test. We found both the series are stationary.

The ARIMA (3, 0, 1) calculates the expected or the predicted values for the total volume series, ARIMA (1, 0, 1) determines the expected/fitted value for the open interest.

<Insert figure 2 about here>

Table 4 reports the descriptive statistics of expected volume; unexpected volume; expected open interest; and the unexpected open interest of the Nifty 50 futures contract. The mean (median) of expected volume (open interest) is 5.9051(5.9152) and 7.2395(7.2459) with standard deviation 0.0940(0.1147), while the mean (median) of unexpected volume (open interest) is 0.0004(0.0001) and 0.0073(-0.0018) with standard deviation of 0.1529(0.0850). Both the unexpected series of volume and open interest exhibits negative (-2.373) and positive skewness (2.6545); and kurtosis value of 17.530(13.9974)

<Insert table 3 about here>

3.2 Volatility Estimation

Unlike previous studies (Schwert, 1990; Bessembinder and Seguin ,1993; Daigler and Wiley, 1999), we have adopted generalized autoregressive conditional heteroskedasticity (GARCH) type models over ordinary least square (OLS) regression, inspired by Gulen and Mayhew (2000), as GARCH models was found to be better adaptive to capture the time- varying returns volatility of financial data. For, the current study, we measure volatility using asymmetric GARCH models; we prefer asymmetric GARCH over GARCH model as GARCH is a symmetric volatility model with constraints imposed on the coefficients while estimating the conditional variance in order to avoid violation of non-negativity conditions, whereas in asymmetric GARCH model overcomes such limitations as no such restrictions are imposed and is more responsive to asymmetric returns behaviour while taking into account the leverage effects.

We tested three alternative asymmetric models, the exponential GARCH model (EGARCH) developed by Nelson, (1991); Threshold GARCH or TARCH³ developed by Zakoian (1994); and asymmetric power ARCH model (APARCH) of Ding et al., (1993). We based our analysis on the EGARCH model in consistent with Gulen and Mayhem (2000), and the other two models are part of robustness checks. Lag1 return (returns equation) and variance (variance equation) terms are also part of the conditional mean and variance equations (Lamoureux and Lastrapes, 1990; Daigler and Wiley, 1999). Further, we integrate trading activity variables as a proxy for predicting unobservable information arrivals. The following EGARCH Model specification

$$\begin{aligned}
 r_t &= \gamma_c + \gamma_{lag} r_{t-1} + \varepsilon_t \\
 \text{Log}(\sigma_t^2) &= \\
 \delta_c + \delta_{\varepsilon,1} \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \delta_{\varepsilon,2} \left(\frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right) + \delta_{lag} \text{Log}(\sigma_{t-1}^2) + \delta_{Exp_Vol} \text{Exp_Vol}_t + \delta_{Unexp_Vol} \text{Unexp_Vol}_t + \\
 &\quad \delta_{Exp_OI} \text{Exp_OI}_t + \delta_{Unexp_OI} \text{Unexp_OI}_t
 \end{aligned} \tag{3}$$

The ARCH coefficients of $\delta_{\varepsilon,1}$ and $\delta_{\varepsilon,2}$ denotes the magnitude and possible asymmetric volatility response to unexpected news releases, and the GARCH term ($\delta_{\varepsilon,2}$) indicates the presence and degree of volatility persistence. Here r_t represents natural logarithmic difference in price levels expressed in percentages; r_{t-1} is the first lag returns; d_i is the dummy variable applied for day effect; σ_t^2 is the conditional variance; The ARCH coefficients ($\delta_{\varepsilon,1}, \delta_{\varepsilon,2}$) denotes the magnitude and possible asymmetric volatility response to unexpected news releases, and the GARCH term (δ_{lag}) indicates the presence and degree of volatility persistence. Typically, GARCH models are estimated by the maximum-likelihood method, however for this study, we have used Nelson's

³This model is also called as GJR-GARCH model (Glosten et al., 1993)

(1991) generalised error distribution (GED) about the conditional distribution of the error term for the estimation of asymmetric GARCH models.

3.3 Modelling the Impact of Macroeconomic News Announcement, and Volume on Volatility

The news effect is regressed in the conditional volatility equations to examine the relative impact of macroeconomic announcement surprises. In order to measure the impact of individual announcements on returns, we collected the actual news announcements and the forecasted value of the original series calculated using ARIMA process. Let $A_{k,t}$ denote the actual value of an announcement of type k at time t and $F_{k,t}$ denote the forecasted value. We measure the surprises in news announcement k in this study as the difference between actual value ($A_{k,t}$) and the forecasted value ($F_{k,t}$) i.e.

$$A_{k,t} - F_{k,t} \quad (4)$$

Since units of measurement differ across the news variables, we use standardised surprise element of the news announcements measured as:

$$NS_{k,t} = \frac{A_{k,t} - F_{k,t}}{\sigma_k} \quad (5)$$

where σ_k is the sample standard deviation of the surprise component of the type k announcement, $A_{k,t} - F_{k,t}$. As σ_k is a constant for each news announcement, the standardization measure should not affect the statistical significance of the estimated coefficients and fit of the model.

In summary, our estimated model is:

$$\begin{aligned}
 & \delta_{\text{lag}} \text{Log}(\sigma_{t-1}^2) + \delta_{\text{Exp_Vol}} \text{Exp_Vol}_t + \delta_{\text{Unexp_Vol}} \text{Unexp_Vol}_t + \delta_{\text{Exp_OI}} \text{Exp_OI}_t \\
 & + \delta_{\text{Unexp_OI}} \text{Unexp_OI}_t \\
 & + \sum_{k=1}^m \gamma_{\text{+venews+ve}} \text{NS}_{k,t} + \sum_{k=1}^m \gamma_{\text{-venews-ve}} \text{NS}_{k,t}
 \end{aligned}
 \tag{6}$$

Further, we test the interaction effect of news announcement on volume-volatility relationship by adding the trading activity variables⁴ (TA) and the interaction terms in the above model: The estimated model is

$$\begin{aligned}
 & \text{Log}(\sigma_t^2) \\
 & = \delta_c + \delta_{\varepsilon,1} \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \delta_{\varepsilon,2} \left(\frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right) + \delta_{\text{lag}} \text{Log}(\sigma_{t-1}^2) + \delta_{\text{Exp_Vol}} \text{Exp_Vol}_t \\
 & + \delta_{\text{Unexp_Vol}} \text{Unexp_Vol}_t + \delta_{\text{Exp_OI}} \text{Exp_OI}_t + \delta_{\text{Unexp_OI}} \text{Unexp_OI}_t \\
 & + \sum_{k=1}^m \gamma_{\text{+venews+ve}} \text{NS}_{k,t} + \sum_{k=1}^m \gamma_{\text{-venews-ve}} \text{NS}_{k,t} + \\
 & \sum_{l=1}^p \varphi_{\text{+ve moderation}} \text{TA}_{l,t}^{\text{exp,unexp}} + \sum_{l=1}^p \varphi_{\text{-ve moderation}} \text{TA}_{l,t}^{\text{exp,unexp}}
 \end{aligned}
 \tag{7}$$

<Insert figure 3 about here>

⁴ TA: Trading activity variable(TA) includes expected and unexpected total volume and open interest position

4. Empirical Results

Table 5 presents the estimates of the conditional volatility with the trading activity variables in the conditional volatility equation. The ARCH coefficients of $\delta_{\varepsilon,1}$ and $\delta_{\varepsilon,2}$ denotes the magnitude and possible asymmetric volatility response to unexpected news releases, both the ARCH coefficients are not statistically significant, while the GARCH term ($\delta_{\varepsilon,2}$) indicates the presence and degree of volatility persistence. We found that GARCH effects are not significant when trading activity variables are included in estimated model. These results are in contrast to those of the findings of Bessembinder and Seguin (1993), thus supporting Lamoureux and Lastrapes (1989), who find similar results that GARCH effects on equity spot volatility are insignificant when volume is included in the model specification.

Table 5 also reports the results of the estimated models, and the magnitude and statistical significance show that the expected and unexpected trading volume has heterogeneous effect on volatility. The estimated coefficient of expected trading volume is positive (2.1194) and statistically significant 1% level, thus finding support to hypothesis 1, while the estimated coefficient of unexpected trading volume is positive (4.6222) and significant at 1%, thus finding support to hypothesis 2. Furthermore, the coefficient of unexpected trading volume is larger than expected volume, indicating that unexpected trading volume affects returns volatility more than the expected trading volume; these results are in consistent with the findings of Bessembinder and Seguin (1993), Watanabe (2001), Frino et al. (2012) and Chen and Tai (2014).

The coefficient estimates for expected open interest are negative (-0.0092) and statistically insignificant, thus finding partial support to hypothesis 3. Thus the results support partially to the findings of Bessembinder and Seguin (1993) and Watanabe (2001), note that negative and significant relationship between expected open interest and volatility, which is in consistent with market depth theory i.e. expected open interest is affiliated to the amount of capital in

the market which enhances liquidity/market depth, thus mitigating volatility shocks. The coefficient estimates of unexpected open interest is positive (3.0810) and statistically significant at 1% level, thus finding support to hypothesis 3; these results differ from those of Bessembinder and Seguin (1993) and Watanbe (2001). This study reports Ljung –Box statistics for the 10th order serial correlation for the residuals. The Ljung-Box *Q statistics* for the residual are insignificant, revealing negative evidences of serial autocorrelation in the index futures contract.

The results of Table 6 provides useful insights into the relationship between news announcements and returns volatility, among the news announcements dividend yield, exchange rate, put-call ratio, broad money have significant impact on returns volatility. And both negative and positive news surprises have significant impacts, but the magnitudes of their effects are statistically different from each other (Hypothesis 5), that is $\gamma_{+ve\text{news}} \neq \gamma_{-ve\text{news}}$ (Wald's test was performed to check the equality of both the coefficients). Further, only negative or positive news surprises are found statistically significant, which indicates the presence of an asymmetric response to news announcements, exception being dividend yield and put-call ratio, where both the types news announcements were found significant.

The estimated coefficient of negative/positive news surprises of dividend yield was statistically significant at 1% level, while the estimated coefficient of positive exchange rate was significant at statistical level of 1%, whereas coefficient of negative/positive news surprises of sentiment indicator was found statistically significant at 5% and 1 % significance level. Lastly, among the other announcements broad money (M3) was significant at 5% level, thus finding support to hypothesis H5.

Table 6 reports the asymmetric impact of news on returns volatility i.e. both negative and positive news surprises affect realized volatility and that negative news surprises have generally

larger impact than positive news surprises. The estimated results for dividend yield in absolute term demonstrate that negative surprise increase volatility more than the positive news surprise ($\gamma_{+venews} = 1.3596$, $\gamma_{-venews} = 1.3335$) while similar results was also reported for PCR, measure for sentiment indicator (shifts in sentiment are negatively correlated with volatility, that is volatility increases (decreases) when sentiments become more bearish/bullish) whereas in case M3 only negative news announcements affects returns volatility. Furthermore only one positive news surprises – exchange rate – significantly increases volatility. Thus, the empirical results partially support our hypothesis H6.

In addition, we examined the interaction effect of news surprises on trading volume and volatility relationship, the results are reported in table 7, and we find that, news announcements of dividend yield, GDP, and PCR moderates the relationship between volume and volatility through trading activity variables of expected and unexpected trading volume and open interest, thus supporting hypothesis 7. The estimated coefficients supports that news announcements are stronger signals of information flow against trading volume variables.

5. Robustness Checks

In this section, we summarize various other specifications we have tested, but did not report here because the results are found substantially similar to those reported in table 5, 6, and 7. The results of all these tests are available on request. We repeated our analysis using Threshold GARCH (TARCH) model of Zakoian (1994), asymmetric power ARCH model (APARCH) of Ding et al., (1993), by and large, all these specifications have yielded results supporting the same conclusions.

6. Conclusion

This paper examines the effect of news announcements, trading activity variables on returns volatility in futures market. Though the empirical research extensively studied the relationship

among trading activity variables and volatility and effect of news announcement on returns volatility separately in both stock and futures market, however few research have examined the connection between them in emerging markets.

In this study, we undertake a comprehensive examination to understand the relationship between news announcement, notably we rely on 8 different types of news announcements, trading activity variables and returns volatility, by extending the methodology used by Bessembinder and Seguin(1992, 1993), for a selected Index futures contract CNX Nifty 50 in the National Stock Exchange (NSE), India.

Using asymmetric GARCH model and with GED distribution for the period spanning 5 years from January, 2012 to December, 2016, our analysis reveals as found for the developed markets, unexpected trading activity variable influences volatility more than expected volume of index futures contract.

Further, we undertake comprehensive examination of the response returns volatility and trading activity variables to 8 different types of news announcements, to identify the set of announcements that affects returns volatility, and volume-volatility relationship. Our analysis reveals that a set of news announcements have large impact on returns volatility and trading activity variables.

Finally, our study has bearing on previous research that has attempted to connect different literature of market segments to compare the impact of news announcements on returns volatility and volume –volatility relationship using a comprehensive dataset providing insightful documentation of relatively consistent picture, which appears to be robust to model specification in an emerging market.

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Figure 1: Time series plot of daily closing prices, returns, trading volume and open interest of Nifty index futures contract

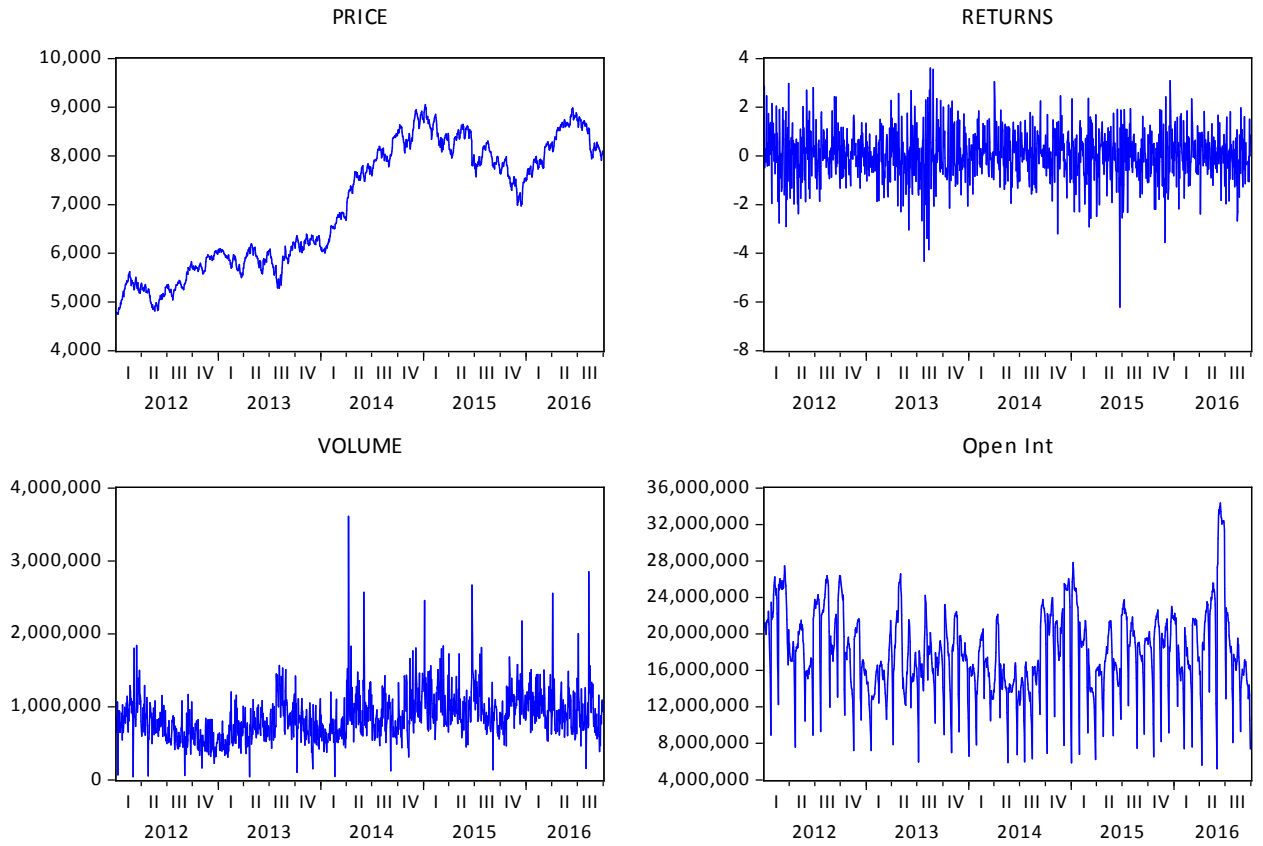


Figure 2: Time series plot of expected and unexpected components of daily trading volume and open interest

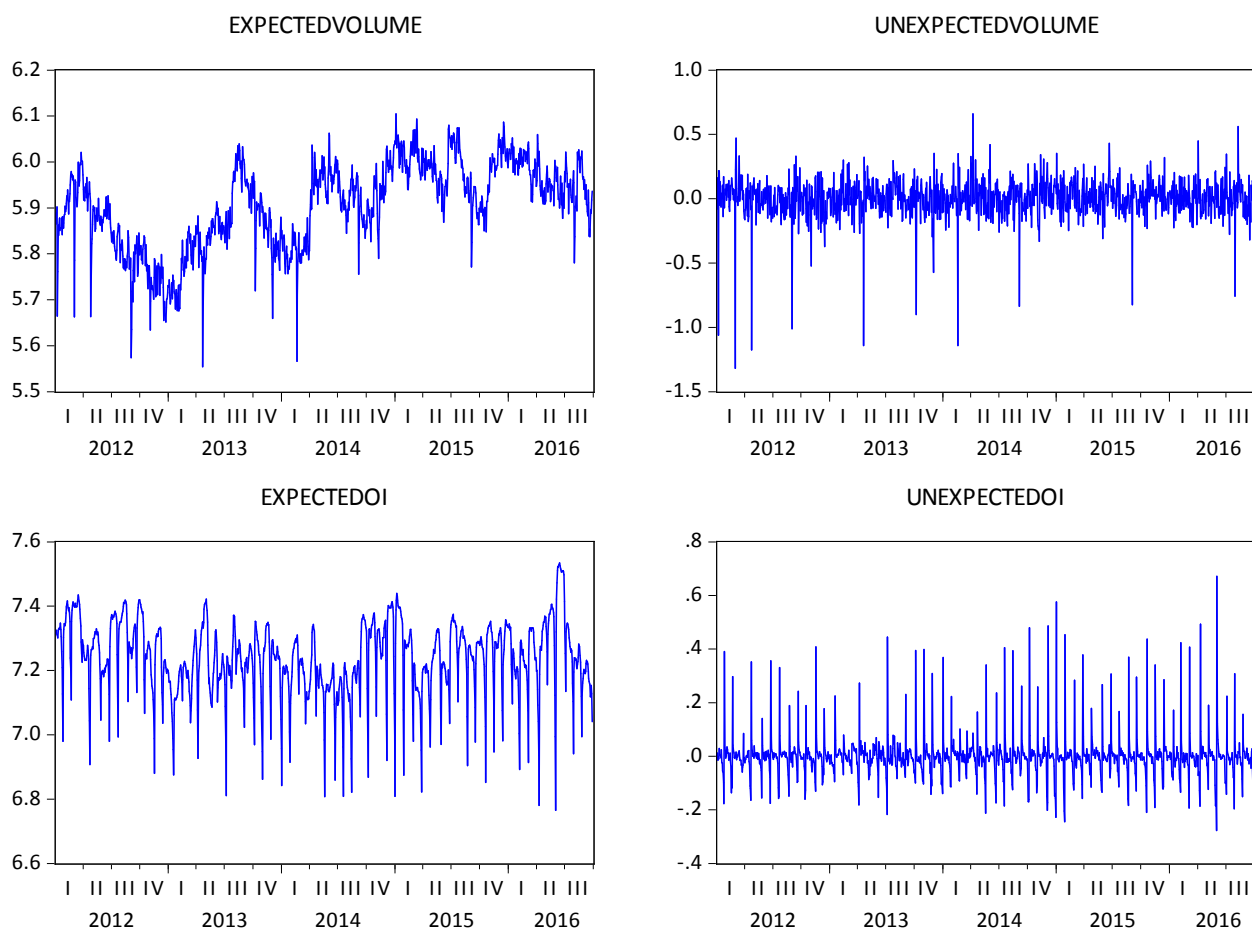


Figure 3: Time series plot of daily news announcements

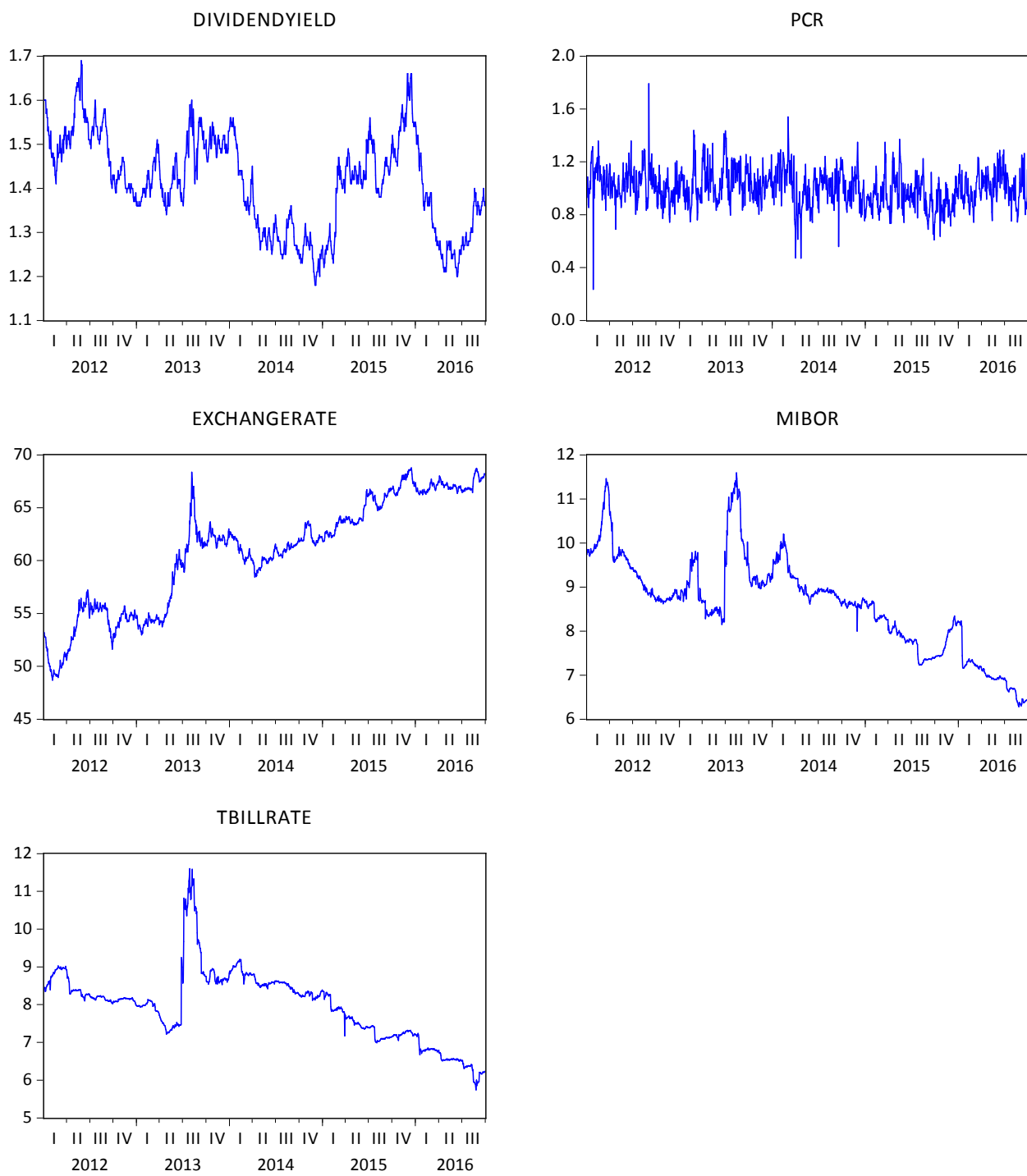


Table 1: Descriptive statistics of daily CNX Nifty 50 Futures contracts

	Returns	Total Volume	Open Interest
Mean	0.00019	5.90511	7.23971
Median	0.00024	5.91538	7.24953
SD	0.00436	0.17932	0.12052
Max	0.01567	6.55777	7.53590
Min	-0.02696	4.63442	6.71768
Skewness	-0.29603	-1.69247	-0.90327
Kurtosis	5.07147	12.86134	4.81357
Jarque–Bera Statistics	239.6203	5611.842	338.2808
	(0.0000)*	(0.0000)*	(0.0000)*
Numberof observations	1239	1239	1239

Note: The future returns are the logarithm returns of nearby month contracts of CNX Nifty 50 futures contracts of National Stock Exchange (NSE), India, estimated for the sample period from January 02, 2012 to December 29, 2016. Jarque-Bera statistics reveals the rejection of normality of the log–return, log-volume, and log-OI process.

*, ** and *** represents significance level at 1%, 5% and 10%.

Table2: Tests on time series property of CNX Nifty 50 Futures Contract

Unit Root Test			
		Panel A	Panel B
Tests Statistics		Augmented-Dickey Test	Fuller Phillip-Perron Test
Return Series		-33.92902 (0.0000)*	-33.90631 (0.0000)*
Volume		-8.66682 (0.0000)*	-31.73867 (0.0000)*
Open -Interest		-13.25865 (0.0000)*	-13.18847 (0.0000)*
Critical Values	1%	-3.43541	
	5%	-2.86366	
	10%	-2.56795	

Note:Unit root problem is diagnosed in return series of CNX Nifty 50 futures contract by Augmented –Dickey Fuller test (ADF) and Phillips –Peron test. Table 2 presents the ADF statistics which is significant at 1% rejecting the null hypothesis of presence of unit root in the return series. Similar results are supported by Phillips Perron (PP) test.

*, ** and *** represents statistical significance level at 1%, 5% and 10%.

Table: 3 Descriptive statistics of original volume, expected volume, unexpected volume, expected open interest and unexpected open interest

Components	Volume			Open Interest		
	Original	Expected	Unexpected	Original	Expected	Unexpected
Mean	5.90511	5.90461	0.00049	7.23971	7.23959	0.00012
Median	5.91538	5.91528	0.00737	7.24953	7.24593	-0.00183
SD	0.17932	0.09402	0.15295	0.12052	0.11476	0.08508
Max	6.55777	6.10481	0.65930	7.53590	7.53452	0.67119
Min	4.63442	5.55408	-1.31749	6.71768	6.76490	-0.27675
Skewness	-1.69247	-0.50887	-2.37395	-0.90327	-0.74641	2.65458
Kurtosis	12.86134	2.78442	17.57037	4.81357	4.50270	13.99742
Number of observation	1239	1239	1239	1239	1239	1239

Note: The descriptive statistics of expected and unexpected components of volume and open interested are computed after decomposition using a ARIMA(p , q) process.

Table: 4 Estimated EGARCH (1, 1) model with volume components

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
δ_c	-12.6978	-2.6119	0.0090	-12.7324	-2.4729	0.0134
$\delta_{\varepsilon,1}$	-0.0707	-1.1442	0.2525	-0.0764	-1.1598	0.2461
$\delta_{\varepsilon,2}$	0.0137	0.3403	0.7336	0.0144	0.3357	0.7371
δ_{lag}	-0.0041	-0.0505	0.9597	-0.0131	-0.1504	0.8804
δ_{Exp_Vol}	2.1194	3.5691	0.0004	2.1218	3.3652	0.0008
δ_{Unexp_Vol}	4.6222	40.1967	0.0000	4.3193	32.1914	0.0000
δ_{Exp_OI}	-0.0092	-0.0217	0.9827	-0.0061	-0.0136	0.9891
δ_{Unexp_OI}	3.0810	5.4177	0.0000	3.1075	5.0954	0.0000
Adjusted R ²	-0.0007			-0.0006		
Ljung-Box Q (10)	10.846					

Note: The table reports the results based on the following specification:

$$\begin{aligned}
 & \text{Log}(\sigma_t^2) \\
 &= \delta_c + \delta_{\varepsilon,1} \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \delta_{\varepsilon,2} \left(\frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right)^2 + \delta_{lag} \text{Log}(\sigma_{t-1}^2) + \delta_{Exp_Vol} \text{Exp_Vol}_t \\
 &+ \delta_{Unexp_Vol} \text{Unexp_Vol}_t + \delta_{Exp_OI} \text{Exp_OI}_t + \delta_{Unexp_OI} \text{Unexp_OI}_t
 \end{aligned}$$

where Exp_Vol_t is the expected trading volume, Unexp_Vol_t is the measurement of unexpected trading volume, Exp_OI_t is the expected open interest and Unexp_OI_t is the measure of unexpected open interest.

Table: 5 Estimated EGARCH (1, 1) model with volume and surprises in news announcements components

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
δ_c	-4.1014	-0.7120	0.4764	-1.8872	-0.6340	0.5260
$\delta_{\varepsilon,1}$	-0.0967	-0.8966	0.3699	-0.1504	-2.3198	0.0203
$\delta_{\varepsilon,2}$	-0.0548	-0.9349	0.3498	-0.0783	-2.3318	0.0197
δ_{lag}	-0.0442	-1.6387	0.1013	-0.0117	-0.7952	0.4265
δ_{Exp_Vol}	0.9129	1.3650	0.1722	0.8717	2.3531	0.0186
δ_{Unexp_Vol}	2.0532	8.1883	0.0000	1.8728	12.1973	0.0000
δ_{Exp_OI}	-0.4539	-0.7830	0.4336	-0.7064	-2.2228	0.0262
δ_{Unexp_OI}	1.9822	2.4505	0.0143	2.0676	3.1980	0.0001
$Y_{-venews,1}^t$	-1.6343	-11.3294	0.0000	-1.3596	-13.2352	0.0000
$Y_{+venews,1}$	1.5756	13.5133	0.0000	1.3335	15.7347	0.0000
$Y_{-venews,2}$	0.0249	0.2016	0.8402	-0.0151	-0.2013	0.8404
$Y_{+venews,2}$	0.1728	2.0577	0.0396	0.2622	5.5478	0.0000
$Y_{-venews,3}$	-0.0523	-0.5845	0.5589	-0.0476	-0.9622	0.3359
$Y_{+venews,3}$	0.0660	0.5041	0.6142	0.0433	0.6172	0.5370
$Y_{-venews,4}$	-0.0874	-0.8564	0.3918	-0.1344	-2.1431	0.0321
$Y_{+venews,4}$	0.1303	1.3431	0.1792	0.2066	3.9497	0.0001
$Y_{-venews,5}$	0.0419	0.3579	0.7204	0.0399	0.6285	0.5296
$Y_{+venews,5}$	0.0206	0.1436	0.8858	0.0124	0.1878	0.8510
$Y_{-venews,6}$	-0.0429	-0.1533	0.8781	-0.2455	-2.0560	0.0398
$Y_{+venews,6}$	-0.0087	-0.0306	0.9756	-0.0154	-0.1135	0.9096

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
$Y_{-venews,7}$	0.0115	0.0406	0.9676	0.0321	0.2168	0.8283
$Y_{+venews,7}$	-0.0634	-0.2538	0.7996	0.0112	0.0718	0.9427
$Y_{-venews,8}$	-1.5767	-0.4329	0.6650	-1.1410	-0.5238	0.6004
$Y_{+venews,8}$	-0.1197	-0.2602	0.7947	-0.1734	-0.6006	0.5481
Adjusted R^2	-0.0005			-0.0032		

Note: The table reports the results based on the following specification:

$$\begin{aligned} & \delta_{lag} \text{Log}(\sigma_{t-1}^2) + \delta_{Exp_Vol} \text{Exp_Vol}_t + \delta_{Unexp_Vol} \text{Unexp_Vol}_t + \delta_{Exp_OI} \text{Exp_OI}_t \\ & + \delta_{Unexp_OI} \text{Unexp_OI}_t + \sum_{k=1}^m \gamma_{+venews+ve} NS_{k,t} + \sum_{k=1}^m \gamma_{-venews-ve} NS_{k,t} \end{aligned}$$

where $+ve NS_{k,t}$ and $-ve NS_{k,t}$ is the measure of positive and negative surprise components of the news announcements.

Table 6 : Estimated EGARCH (1, 1) model with interaction effects

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
δ_c	-4.7068	-0.3098	0.7567	-1.2717	-0.1293	0.8971
$\delta_{\varepsilon,1}$	-0.0911	-0.7368	0.4612	-0.1228	-1.6052	0.1084
$\delta_{\varepsilon,2}$	-0.0183	-0.2713	0.7861	-0.0405	-1.0295	0.3032
δ_{lag}	-0.0426	-1.3049	0.1919	0.0090	0.4865	0.6266
δ_{Exp_Vol}	0.9782	0.5687	0.5695	0.4016	0.3652	0.7149
δ_{Unexp_Vol}	3.3211	5.2837	0.0000	3.1625	7.2370	0.0000
δ_{Exp_OI}	-0.4261	-0.2566	0.7974	-0.3963	-0.3631	0.7165
δ_{Unexp_OI}	3.8143	1.6820	0.0926	4.3567	2.7609	0.0058
$Y_{-venews,1}$	-2.0697	-0.1311	0.8957	-1.7292	-0.1771	0.8594
$Y_{+venews,1}$	4.7655	0.4009	0.6884	-0.6732	-0.0842	0.9328
$Y_{-venews,2}$	-2.8288	-0.1798	0.8573	-2.2999	-0.2578	0.7965
$Y_{+venews,2}$	-2.2069	-0.1787	0.8581	-2.3344	-0.3540	0.7233
$Y_{-venews,3}$	-5.1699	-0.3848	0.7003	-3.2479	-0.3665	0.7139
$Y_{+venews,3}$	0.1339	0.0093	0.9925	1.0179	0.1303	0.8963
$Y_{-venews,4}$	6.1218	0.4756	0.6343	4.0034	0.5539	0.5796
$Y_{+venews,4}$	4.3017	0.3463	0.7291	3.1289	0.3791	0.7046
$Y_{-venews,5}$	1.0638	0.0613	0.9511	-2.2652	-0.2562	0.7978
$Y_{+venews,5}$	0.6123	0.0258	0.9793	-0.2481	-0.2530	0.8003
$Y_{-venews,6}$	3.1536	0.0578	0.9539	-7.8974	-0.2480	0.8041
$Y_{+venews,6}$	-4.5144	-0.0724	0.9422	3.8916	0.1001	0.9202
$Y_{-venews,7}$	33.4505	0.6271	0.5306	21.5334	8.7670	0.0000

Cont...

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
$Y_{+ve news,7}$	-4.5343	-0.0949	0.9244	0.4268	0.01292	0.9897
$Y_{-ve news,8}$	-78.1367	-0.0469	0.9626	-88.4208	-743366.9	0.0000
$Y_{+ve news,8}$	17.1912	0.1771	0.8594	12.3271	0.4786	0.6322
$\varphi_{-ve moderation,1}$	1.0694	0.6279	0.5300	1.0539	0.9417	0.3463
$\varphi_{+ve moderation,1}$	-2.0316	-1.3271	0.1845	-1.1426	-1.0976	0.2724
$\varphi_{-ve moderation,2}$	-0.3118	-0.1885	0.8505	-0.5459	-0.5091	0.6106
$\varphi_{+ve moderation,2}$	0.7505	0.5436	0.5867	1.2289	1.3877	0.1652
$\varphi_{-ve moderation,3}$	-0.2134	-0.1160	0.9076	-0.2111	-0.1865	0.8520
$\varphi_{+ve moderation,3}$	0.2053	0.0993	0.9209	0.0918	0.0823	0.9344
$\varphi_{-ve moderation,4}$	-0.5568	-0.3698	0.7115	-0.6781	-0.7669	0.4431
$\varphi_{+ve moderation,4}$	0.1370	0.0954	0.9240	0.3424	0.3734	0.7088
$\varphi_{-ve moderation,5}$	0.4753	0.2113	0.8326	0.4945	0.3985	0.6902
$\varphi_{+ve moderation,5}$	0.0023	0.0008	0.9993	0.3076	0.5156	0.6061
$\varphi_{-ve moderation,6}$	-0.8910	-0.2313	0.8170	-1.5925	-0.6704	0.5026
$\varphi_{+ve moderation,6}$	-1.9153	-0.2003	0.8412	-2.2599	-0.4589	0.6463
$\varphi_{-ve moderation,7}$	-2.0224	-0.3305	0.7410	-1.0461	-0.4195	0.6748
$\varphi_{+ve moderation,7}$	3.4688	0.5238	0.6004	2.6283	0.6452	0.5187
$\varphi_{-ve moderation,8}$	-40.1539	-0.1002	0.9201	-40.0349	-31062.12	0.0000
$\varphi_{+ve moderation,8}$	-1.0114	-0.0708	0.9435	1.8184	0.2834	0.7769
$\varphi_{-ve moderation,9}$	0.2784	0.1912	0.8484	0.1463	0.1592	0.8735

Cont...

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
$\varphi_{+ve\ moderation,9}$	-0.7174	-0.8027	0.4221	-0.7155	-1.1221	0.2618
$\varphi_{-ve\ moderation,10}$	0.5296	0.3344	0.7380	0.8229	0.7915	0.4286
$\varphi_{+ve\ moderation,10}$	-0.2882	-0.3102	0.7564	-0.4815	-0.7492	0.4537
$\varphi_{-ve\ moderation,11}$	-0.0740	-0.0584	0.9534	0.0061	0.0080	0.9936
$\varphi_{+ve\ moderation,11}$	0.2980	0.2401	0.8102	0.4336	0.5511	0.5815
$\varphi_{-ve\ moderation,12}$	0.8613	1.7898	0.0735	0.7930	2.8857	0.0039
$\varphi_{+ve\ moderation,12}$	-0.4765	-0.9796	0.3273	-0.5520	-1.7346	0.0828
$\varphi_{-ve\ moderation,13}$	0.2373	0.2797	0.7796	-0.0252	-0.0464	0.9630
$\varphi_{+ve\ moderation,13}$	-0.2496	-0.1312	0.8956	0.6214	0.6099	0.5419
$\varphi_{-ve\ moderation,14}$	-1.0473	-0.3147	0.7529	-1.4396	-0.8900	0.3735
$\varphi_{+ve\ moderation,14}$	4.8804	0.7464	0.4554	4.6439	1.1157	0.2645
$\varphi_{-ve\ moderation,15}$	2.1356	0.5051	0.6134	1.1305	0.4079	0.6833
$\varphi_{+ve\ moderation,15}$	1.8516	0.3284	0.7426	1.7709	0.5162	0.6057
$\varphi_{-ve\ moderation,16}$	-3.5640	-0.0570	0.9545	-8.0724	-0.1534	0.8780
$\varphi_{+ve\ moderation,16}$	1.1041	0.0940	0.9251	4.3117	0.6615	0.5082
$\varphi_{-ve\ moderation,17}$	-0.8254	-0.4512	0.6518	-0.8120	-0.7544	0.4506
$\varphi_{+ve\ moderation,17}$	1.2255	1.0044	0.3152	1.2048	1.4528	0.1463
$\varphi_{-ve\ moderation,18}$	0.6452	0.3914	0.6955	0.7566	0.8704	0.3840
$\varphi_{+ve\ moderation,18}$	-0.2826	-0.2105	0.8332	-0.6417	-0.8269	0.4083
$\varphi_{-ve\ moderation,19}$	0.8863	0.6101	0.5418	0.6205	0.6254	0.5317
$\varphi_{+ve\ moderation,19}$	-0.1840	-0.1072	0.9146	-0.2227	-0.2270	0.8204

Cont...

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
$\Phi_{-ve\ moderation,20}$	-0.4004	-0.3140	0.7534	-0.0079	-0.0104	0.9916
$\Phi_{+ve\ moderation,20}$	-0.6978	-0.4857	0.6271	-0.6950	-0.8312	0.4058
$\Phi_{-ve\ moderation,21}$	-0.5387	-0.2808	0.7788	-0.0936	-0.0926	0.9262
$\Phi_{+ve\ moderation,21}$	-0.0851	-0.0371	0.9703	-0.2151	-0.6135	0.5395
$\Phi_{-ve\ moderation,22}$	0.2775	0.0404	0.9677	2.3666	0.6058	0.5446
$\Phi_{+ve\ moderation,22}$	2.1745	0.2232	0.8234	1.2823	0.2365	0.8130
$\Phi_{-ve\ moderation,23}$	-2.9460	-0.5906	0.5547	-2.0905	-0.9778	0.3281
$\Phi_{+ve\ moderation,23}$	-2.2258	-0.3533	0.7236	-2.2102	-0.5179	0.6045
$\Phi_{-ve\ moderation,24}$	43.7106	0.1449	0.8847	45.3185	25.5144	0.0000
$\Phi_{+ve\ moderation,24}$	-1.6114	-0.2112	0.8327	-3.2589	-0.6299	0.5287
$\Phi_{-ve\ moderation,25}$	1.1370	0.4913	0.6232	1.1838	0.8202	0.4121
$\Phi_{+ve\ moderation,25}$	-3.9615	-1.7800	0.0751	-3.8033	-2.6676	0.0076
$\Phi_{-ve\ moderation,26}$	0.8437	0.2897	0.7720	1.4339	0.9032	0.3664
$\Phi_{+ve\ moderation,26}$	0.0820	0.0357	0.9715	-1.4383	-0.8618	0.3888
$\Phi_{-ve\ moderation,27}$	0.0311	0.0090	0.9927	-1.4442	-0.6499	0.5157
$\Phi_{+ve\ moderation,27}$	-0.6020	-0.2259	0.8212	-0.3558	-0.1973	0.8436
$\Phi_{-ve\ moderation,28}$	0.2799	0.1804	0.8568	0.3094	0.3327	0.7393
$\Phi_{+ve\ moderation,28}$	0.0395	0.0250	0.9800	0.0562	0.0535	0.9573
$\Phi_{-ve\ moderation,29}$	-0.3432	-0.1459	0.8839	0.2434	0.1810	0.8564
$\Phi_{+ve\ moderation,29}$	0.2171	0.0645	0.9486	0.8798	0.4464	0.6552
$\Phi_{-ve\ moderation,30}$	-1.0487	-0.1123	0.9105	-4.9844	-0.8019	0.4226

Cont...

Coefficients	MLE			GED		
	Values	Z Statistics	p Values	Values	Z Statistics	p Values
$\varphi_{+ve\ moderation,30}$	1.6101	0.0596	0.9524	3.7260	0.1594	0.8733
$\varphi_{-ve\ moderation,31}$	14.8159	0.3264	0.7441	9.4113	0.3241	0.7458
$\varphi_{+ve\ moderation,31}$	-1.3872	-0.1383	0.8900	-2.0758	-0.3641	0.7157
$\varphi_{-ve\ moderation,32}$	37.7640	0.1523	0.8789	36.0364	0.4315	0.6660
$\varphi_{+ve\ moderation,32}$	0.6119	0.0522	0.9584	-0.7921	-0.1335	0.8937
Adjusted R^2	-0.0009			-0.0074		

Note: The table reports the results based on the following specification:

$\text{Log}(\sigma_t^2)$

$$\begin{aligned}
&= \delta_c + \delta_{\varepsilon,1} \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \delta_{\varepsilon,2} \left(\frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right) + \delta_{\text{lag}} \text{Log}(\sigma_{t-1}^2) + \delta_{\text{Exp_Vol}} \text{Exp_Vol}_t \\
&+ \delta_{\text{Unexp_Vol}} \text{Unexp_Vol}_t + \delta_{\text{Exp_OI}} \text{Exp_OI}_t + \delta_{\text{Unexp_OI}} \text{Unexp_OI}_t \\
&+ \sum_{k=1}^m \gamma_{+ve\ news} +ve\ NS_{k,t} + \sum_{k=1}^m \gamma_{-ve\ news} -ve\ NS_{k,t} \\
&+ \sum_{l=1}^p \varphi_{+ve\ moderation} TA_{l,t}^{\text{exp,unexp}} + \sum_{l=1}^p \varphi_{-ve\ moderation} TA_{l,t}^{\text{exp,unexp}}
\end{aligned}$$

where $TA_{l,t}^{\text{exp,unexp}}$ and $TA_{l,t}^{\text{exp,unexp}}$ are the series of expected and unexpected components of trading volume and open interest